

PALLETPILOT · CALCULATION

PalletPilot sizing calculations

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	PLP-CAL-001	Version	0.6
Status	Draft	Date	2026-10-02
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/palletpilot		
Source	docs/04-calcs/01-sizing.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.6	2026-10-02	Amish Chadha	R15a status against the 45 kg prototype limit decided by Amish on 2026-10-02; totals updated; no figures changed
0.5	2026-10-01	Amish Chadha	Steering range paragraph records Amish's decision (PLP-DDR-003 A1) and the TRL 4 aisle test gate; no number changed, script not rerun
0.4	2026-10-01	Amish Chadha	Rerun on the constructable model (PLP-DDR-003); mass from modelled volumes, release mechanism [G4] and steering range [G5]; budget treated as a value-engineering target
0.3	2026-09-26	Amish Chadha	Budget approved by Amish (\$1,550 to \$1,610, PLP-DDR-002); script rerun; R17 not met to met
0.2	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.1	2026-09-25	Amish Chadha	First issue for TRL 3 (mass, axle loads, traction, torque, steering, layered stopping, UWB error budget, energy, geometry, cost, requirement status)

PalletPilot sizing calculations

On paper, PalletPilot meets eleven of its eighteen requirement lines, misses one, has two at risk, three cannot be verified at TRL 3, and cost is reported against its value-engineering target. The drive, ramp, bumper, energy, charging, parking, handle clearance, length and hand-push targets are met. The miss is follow-mode bearing accuracy (R5, about $\pm 16^\circ$ against $\pm 10^\circ$); the kit mass (R15, 44.4 kg) meets the 45 kg prototype limit set by Amish on 2026-10-02 (PLP-DEC-001) and misses the 40 kg goal. Value-engineering target: USD 1,610. Estimated cost of the constructable design: USD 1,690 (USD 80 over the target). Version 0.4 reruns the note on the constructable model of PLP-DDR-003: made parts are weighed from their modelled volumes, the release lever is sized from the cam and arm geometry, and the steering range with the kit fitted is checked. Version 0.2 applies Amish's 2026-09-25 decisions in PLP-DDR-002: the bumper-only speed is limited to 0.15 m/s (R8 now met), a second main contactor is added to the stop chain (+\$30, +0.3 kg), and the mass and cost overruns are accepted for now with no budget change. Version 0.3 applies only the budget change to \$1,610 (PLP-DDR-002); no design number changed. The approved lidar layer stops the loaded truck from 0.6 m/s in 0.42 to 0.76 m, inside a 0.86 m protective field that stays clear of the operator it follows. That is the core of the layered-stopping case, but the sensor is not safety-rated, so R7 stays at risk. Every number in this note is printed by `docs/04-calcs/sizing.py`; the tag in brackets, for example [F3], is the line of the script's output that carries it.

Safety: These are first-principles estimates for a paper proof of concept of moving machinery that carries up to 1.6 t near people's feet, with a 512 Wh lithium pack. They do not replace brake, stop-chain, detection or electrical tests, and they give no basis for ISO 3691-4 or ISO 13849-1 claims. Nothing may be built or run on the strength of this note. See PLP-PRC-001, Safety.

■ Scope and method

The note checks every requirement in PLP-REQ-001 v0.6 against the design in PLP-PRC-001 v0.6, as decided in PLP-DDR-001 and PLP-DDR-002 and made constructable in PLP-DDR-003. The script imports `PARAMS` and `derived()` from `cad/src/model.py`, so the geometry here matches the STEP files and drawing PLP-DWG-001, and it reads the BOM total from `bom/bom.csv`. Run it from the repo root with `python docs/04-calcs/sizing.py`.

■ Assumptions

Table 1. Main assumptions.

Area	Assumption	Basis
Loads	1,000 kg design pallet, 1,500 kg maximum; donor jack 64 kg; pallet center 615 mm from the frame head; 45 % of the jack's mass on the steer axle	PLP-REQ-001; 48 x 40 in pallet centered on 48 in forks
Floor	Rolling resistance 1.2 %, breakaway 2.5 %; drive tread friction 0.5 on dusty sealed concrete, 0.4 as a low case	Polyurethane wheels on concrete; to be measured
Drive	Two 200 mm hub motors; spring preload 1.5 kN on the pair (raised from 1.1 kN at TRL 2); specified minimums of 12 N·m continuous, 30 N·m peak and 20 N·m spring-brake torque per motor	Motor class data: 20/40 N·m (UU Motor SVB8S) and 40/80 N·m (ZLTECH 8 in); brake torque to confirm
Acceleration	0.3 m/s ² level, 0.1 m/s ² on the 2 % ramp, 0.2 m/s ² at 1,500 kg	Controller settings
Stopping	Lidar layer: controlled stop by the motor driver, delay 0.25 s (two 10 Hz scans plus processing). Bumper and emergency stops: safety relay drops the two series contactors and the spring brakes act, delay 0.10 s. Tag loss: three missed frames at 20 Hz plus command, 0.20 s	RPLIDAR C1 scans at 8 to 12 Hz; relay and brake times typical of the part class
Follow geometry	Gap 1.5 m ± 0.3 m; tag worn at the front of the body, 0.15 m from the back of the legs	R5
UWB	Range noise 0.10 m (1 σ) per anchor; 20 Hz, averaged over 0.5 s (10 samples); correlated error 0.03 m per range from multipath and body shadowing; R5's ±10° read as a 2 σ bound	Qorvo DWM3000 about 10 cm precision; correlated error is an assumption
Energy	60 moves of 40 m loaded and 40 m empty, four starts each way to 1.2 m/s; no regeneration; motor and driver 70 %; controls, UWB, lidar and lights 9.2 W; 80 % usable depth; cell 95 %, charger 88 %	PLP-REQ-001 design duty
Mass	Hub motor 7.0 kg each (class range 3 to 8.5 kg); made steel and aluminium parts weighed from their modelled volumes; 2 mm aluminium enclosure; 8S6P 26650 pack	Motor class data above; <code>model.masses()</code>

■ Mass and axle loads

The kit weighs about 44.4 kg on the truck [B1], 4.4 kg over R15's 40 kg goal and 0.6 kg under its 45 kg prototype limit (decided 2026-10-02). The two hub motors (14 kg) and the drive module (10.6 kg: top plate, lower jaw, drive arms and pivot pin) are 55 % of it; the release mechanism adds 3.3 kg [B0]. Making the design constructable added 2.4 kg (PLP-DDR-003). The design total is 1,108 kg, and 1,608 kg with a 1,500 kg pallet [B2].

The steer axle carries 41.5 % of the pallet. The yoke's ground load is 4.79 kN at 1,000 kg and 6.83 kN at 1,500 kg [C1]. With 1.5 kN of spring preload on the drive wheels, the steer wheels keep 3.29 kN at the design load [C2]. With the jack empty, the yoke load (0.72 kN) is below the preload, so the steer wheels lift and the drive wheels carry the yoke; traction is then 0.34 g, which is ample [C3]. The preload springs are stiff (about 100 N/mm, 4.2 mm of preload compression), so the springs extend only about 2 mm before the load balances and the steer wheels lift only about 1.2 mm.

The kit hangs on the yoke, so it adds no vertical load to the jack's steering bearing, which carries 4.36 kN at 1,000 kg against 10.47 kN at the donor's rating. The bearing does take up to 750 N of horizontal drive force [C4], about 2.8 times the breakaway pull of a person, which makes bearing condition a donor survey item.

■ Forces, traction and torque

Table 2. Drive cases, design total unless stated [F1 to F4].

Case	Force	Torque per motor	Traction margin, friction 0.5 / 0.4
Level start (breakaway)	272 N	13.6 N·m	2.76 / 2.21
Level, accelerate 0.3 m/s ²	463 N	23.2 N·m	1.62 / 1.30
2 % ramp start (breakaway plus grade)	489 N	24.5 N·m	1.53 / 1.23
2 % ramp, accelerate 0.1 m/s ²	459 N	22.9 N·m	1.63 / 1.31
1,500 kg level, accelerate 0.2 m/s ²	511 N	25.6 N·m	1.47 / 1.17
1,500 kg, 2 % ramp start	710 N	35.5 N·m	1.06 / 0.85

Raising the preload from 1.1 to 1.5 kN lifts the available traction from 550 to 750 N [F2], which turns R3 from at risk into met with a margin of 1.53. The worst case inside R2 and R3 needs 25.6 N·m per motor against the specified 30 N·m peak [F4]. A 1,500 kg ramp start needs 35.5 N·m and has almost no traction margin, so **the 1,500 kg rating applies to level floors only**. The TRL 2 motor class (about 16 N·m peak) is too weak; the BOM now calls for a 30 N·m peak class motor.

Cruising at 1.2 m/s loaded takes 6.5 N·m per motor against 12 N·m continuous, at 115 rpm [F5]. Peak wheel power is 556 W while accelerating, drawing about 31 A from the pack, about 16 A per driver channel [F6]. The 2 x 15 A continuous, 30 A peak driver and the 50 A BMS cover this.

Steering finding. The drive axle sits 180 mm from the steering axis, toward the handle end. The motors' differential push can produce about 98 N·m about the steering axis, but swinging the yoke at standstill scrubs the drive wheels sideways and needs about 135 N·m [F7]. The motors therefore cannot steer at standstill; follow mode must steer only while rolling, when the wheels follow arcs and the scrub disappears. With the drive end leading in follow mode, the yoke trails behind the drive axle, which is the stable arrangement. In walk mode, standstill steering by hand gets heavier by about 112 N at the grip [F8] until the truck rolls. No requirement covers steering effort; see REVIEW.md.

■ Layered stopping

Braking force is limited by motor torque in a controlled stop (600 N) and by the spring brakes in an emergency stop (400 N) [S1]. Rolling resistance adds to both.

Table 3. Stopping distances [S2].

Case	Deceleration	Stop	Kinetic energy
Walk 1.2 m/s, controlled, level	0.66 m/s ²	1.21 m	798 J
Walk 1.2 m/s, emergency stop, level	0.48 m/s ²	1.62 m	798 J
Walk 1.2 m/s, emergency stop, 2 % downgrade	0.28 m/s ²	2.67 m	798 J
Follow 0.6 m/s, lidar stop, level	0.66 m/s ²	0.42 m	200 J
Follow 0.6 m/s, lidar stop, 2 % downgrade	0.46 m/s ²	0.54 m	200 J
Follow 0.6 m/s, lidar stop, 1,500 kg, 2 % downgrade	0.29 m/s ²	0.76 m	290 J
Follow 0.6 m/s, bumper only, level	0.48 m/s ²	0.44 m	200 J
Creep 0.3 m/s, bumper only, level	0.48 m/s ²	0.12 m	50 J
0.15 m/s bumper-only limit, level	0.48 m/s ²	0.039 m	12 J
0.2 m/s, bumper only, level (v0.1 limit, for comparison)	0.48 m/s ²	0.062 m	22 J

Lidar layer (approved option 1A). A 2D lidar sits just inside the bumper face with its scan plane 200 mm above the floor, above the bumper hoop and the UWB anchors [G3]. A protective field of 0.86 m ahead of the bumper face (the worst lidar stop plus 0.1 m) [S4] covers every follow case in Table 3. It is 885 mm wide, the jack width plus 100 mm each side [S5]. The operator's legs are at least 1.05 m from the bumper at the minimum gap, so the field stays clear of the person being followed by 0.19 m. A slow-down field beyond it should cut speed before a stop is needed. The distances fit on paper, but the sensor is not safety-rated, its range on black targets falls to 6 m, and its detection of dark or shiny clothing at shin height is unproven. **R7 is therefore at risk, not met**, and no ISO 3691-4 claim can be made.

Bumper layer. The chain acts in 0.10 s, which meets R8's 100 ms. The speed that stops within the edge's 40 mm travel is 0.15 m/s [S3]. Amish decided on 2026-09-25 to limit bumper-only operation to 0.15 m/s (PLP-DDR-002, O4) rather than fit a longer-travel edge. At that limit the truck travels 39 mm after contact, inside the 40 mm travel, so **R8 is met on paper**; at the former 0.2 m/s it would need 62 mm. Because the lidar layer is untested, follow mode also runs at 0.15 m/s or less until that layer is built and tested. The stop chain now drops two main contactors in series, one per safety relay channel (PLP-DDR-002, O6); the performance level is still to be calculated, so R9 stays at risk.

Tag loss. The stop command follows 0.20 s after the last good frame, inside R6's 0.3 s [S6]; the firmware does not exist, so R6 cannot be verified at TRL 3.

Parking. Holding the design load on 2 % needs 21.7 N·m in total; the specified brakes give 40 N·m, a margin of 1.84 [S7].

■ Follow-me tracking: UWB error budget

Two anchors 450 mm apart give a raw bearing error of 18.0° (1 σ) [U1]. Averaging 10 samples brings the random part to 5.7°, but the correlated part (5.4°) does not average out; the total is 7.8° (1 σ), or about $\pm 16^\circ$ at 2 σ against $\pm 10^\circ$ [U2]. Meeting R5 with this filter would need a 706 mm baseline, wider than the 685 mm jack [U3]. At the 1.5 m gap, $\pm 16^\circ$ is about 0.42 m of lateral error [U4]. The gap itself is held to about ± 0.06 m, well within ± 0.3 m. **R5 is not**

met on paper. The routes are phase-difference angle-of-arrival UWB modules, fusing the lidar's track of the operator's legs with the UWB range, or accepting a wider bearing band; each needs a bench measurement, which is TRL 4 work.

■ Energy and charging

Table 4. Energy per shift [E1 to E4].

Quantity	Value
Work at the wheels	154 Wh (2.57 Wh per move)
Motor driver input	220 Wh
Controls, UWB, lidar and lights	74 Wh
Energy from the pack	293 Wh of 410 Wh usable (28 % left)
Energy from the wall	351 Wh
Charge time at 5 A	4.5 h

R12 (20 % left) and R13 (5 h) are met. The lidar adds about 10 Wh per shift.

■ Geometry

The TRL 2 enclosure sat above the handle pivot, so the lowered handle would have hit its lid at about 51° from vertical, inside the 20° to 70° walk band. The TRL 3 model lowers the enclosure top to 320 mm, below the 350 mm pivot, and moves the beacon to the tiller head and the second emergency stop to the enclosure's rear face. The handle now clears the enclosure by 40 mm at 70° and by 14 mm when fully lowered [G1]. The pivot height varies between jacks, so this is a donor survey item.

Moving the drive axle 40 mm closer to the steering axis and widening the drive track to 260 mm, so the drive wheels pass beside the steer wheels, brings the added length at floor level to 290 mm, within R15's 300 mm [G2]. The truck is 1,710 mm long at floor level against 1,420 mm bare.

Release (PLP-DDR-003). Each drive arm pivots behind its wheel; its spring acts at the arm's front end, 1.77 times as far from the pivot as the wheel, so 423 N per spring gives 750 N per wheel. The 400 mm lever turns a cam shaft a quarter turn through a link; each 22 mm eccentric cam lifts its spring saddle 22 mm, the springs unload after 4.2 mm and slotted straps then lift the arms, so the drive wheels clear the floor by about 10 mm. The peak effort at the grip is about 58 N, half the 117 N estimated for the concept's cam, and with the drive wheels lifted the unpowered push force rises only 4.2 % [G4].

Steering range. With the kit fitted, the drive end swings into the jack's frame head when the handle is turned far. On the reference donor (frame head 100 mm ahead of the steering axis) the first kit part would reach it at 44°; two rubber stops meet it at 40° each way [G5], against about 90° for a bare jack. That gives a turning radius of roughly 1.5 m about the load wheels (on the reference donor the load rollers are 1,130 mm from the steering axis). Amish accepted this range for the prototype on 2026-10-01 (PLP-DDR-003, A1; PLP-DEC-001); the TRL 4 aisle test confirms it and is a go or no-go gate for the drive layout.

■ Cost

Value-engineering target: USD 1,610 (`budget_usd` , a hypothetical control target, not a limit; Amish, 2026-10-01). Estimated cost of the constructable design: USD 1,690 over 19 BOM lines (USD 80 over the target, 5.0 %) [K1]. The design for construction added USD 80: the drive module with its clamp, arms and pivot (\$90 to \$130), the release mechanism (\$20 to \$45), the enclosure feet (\$5) and the steering stops (\$10, new line 19). The motor price is still the least certain line. Savings worth trying are listed in PLP-DEC-001.

■ Results against requirements

Table 5. Requirement status. Values from the [R] lines of the script.

ID	Target	Value	Status
R1	Bolts to common 27 x 48 in jacks; fit 2 h, remove 1 h	Clamp to the yoke, no drilling; donor geometry assumed	Not verifiable at TRL 3
R2	Move 1,000 kg; 1,500 kg at 0.8 m/s or less	23.2 and 25.6 N·m per motor of 30 N·m; traction 750 N	Met (1,500 kg on level floors only)
R3	Start, climb and hold 1,000 kg on 2 %	Start needs 489 N of 750 N (margin 1.53, 1.23 at friction 0.4); hold margin 1.84	Met
R4	Walk 1.2, forks first 0.8, creep 0.3, follow 0.6 m/s	Controller limits	Met
R5	Gap ± 0.3 m, bearing $\pm 10^\circ$	Gap ± 0.06 m; bearing $\pm 16^\circ$ (2σ)	Not met
R6	Stop within 0.3 s of losing the tag	Timing budget 0.20 s; no firmware	Not verifiable at TRL 3
R7	Detect a person and stop without contact from 0.6 m/s	Stop 0.42 to 0.76 m inside a 0.86 m field; sensor not safety-rated	At risk
R8	Bumper stop in 100 ms; bumper-only 0.15 m/s or less	0.10 s; 0.15 m/s stops in 39 mm of 40 mm travel	Met
R9	Hardwired stops through a relay and contactors; PL d	Dual-channel relay, two contactors in series; PL not calculated	At risk
R10	Brakes hold 2 % with power off	40 N·m specified against 21.7 N·m needed	Met (datasheet to confirm)
R11	Walk drive only between 20° and 70° ; belly reverse	Handle clears the enclosure by 40 mm at 70° , 14 mm at 90°	Met
R12	Shift with 20 % or more left	293 of 410 Wh; 28 % left	Met
R13	Charge in 5 h or less	4.5 h	Met
R14	Release in 10 s; push force +10 % or less	+4.2 %; lever 58 N; wheels lift 10 mm	Met
R15a	Kit mass 45 kg or less for the prototype (40 kg goal)	44.4 kg	Met for the prototype (goal not met)
R15b	Added length 300 mm or less	290 mm	Met
R16	0 to 40 °C, IP54, no charging below 0 °C	By specification of bought parts	Not verifiable at TRL 3
R17	Kit parts against the \$1,610 value-engineering target	\$1,690	Over the target by \$80

Totals: 11 met, 1 not met (R5), 2 at risk, 3 not verifiable at TRL 3, and R17 over its value-engineering target by \$80 [R0].

■ Checks against earlier figures

Table 6. TRL 2 figures checked against this note.

TRL 2 figure (PLP-PRC-001 v0.2)	TRL 3 value	Change
Kit mass about 37 kg	44.4 kg (42.0 kg in v0.3)	Stronger, heavier motors; lidar; second contactor (v0.2); parts added for construction (v0.4); R15 mass not met
Available traction 550 N	750 N	Preload raised to 1.5 kN
Wheel torque 17 N·m per motor	25.6 N·m (30 N·m specified)	Acceleration and 1,500 kg cases added
Stop from 1.2 m/s about 1.56 m	1.21 m controlled, 1.62 m emergency	Braking now split by stop type
Stop from 0.6 m/s about 0.42 m	0.42 m level, up to 0.76 m worst case	Same level figure; worst cases added
Bumper-only safe at about 0.2 m/s	0.15 m/s (now the decided limit)	TRL 2 compared the stop with no travel limit; 0.2 m/s needs 62 mm
Raw bearing error about $\pm 18^\circ$	18.0° (1σ); $\pm 16^\circ$ (2σ) filtered	Confirmed; filtering does not close the gap
Energy 281 Wh of 410 Wh	293 Wh	Heavier kit and the lidar
Wall energy 336 Wh	351 Wh	As above
Added length about 330 mm	290 mm	Drive axle moved, track widened
Kit cost about \$1,425	\$1,690 (\$1,610 in v0.3)	Motors \$60 more in total; lidar \$80; second contactor \$30 (v0.2); parts added for construction \$80 (v0.4)

The docs PLP-PRC-001 and PLP-REQ-001 v0.6, PLP-PRB-001, the README, the BOM notes and the concept media use the values of this version.